CLASS 9 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A student finds a cell with no nucleus, a cell wall and DNA free in the cytoplasm. The cell is: 1

(a) animal **(b)** plant **(c)** bacterial **(d)** fungal

2. After removing a ring of bark, the leaves above remain fresh for some time because: 1

(a) cork is intact **(b)** cambium is intact
(c) phloem is intact **(d)** xylem is intact

3. A student finds a plant with a vascular system, seeds in cones and no flowers. The class is: 1

(a) Gymnosperms **(b)** Pteridophytes **(c)** Bryophytes **(d)** Angiosperms

4. The egg released on 14 March remains fertilisable for about 1 to 2 days. The most fertile days are: 1

(a) 1 to 5 March **(b)** 20 to 25 March **(c)** 28 to 30 March **(d)** 13 to 15 March

5. A wetland is drained for building. Which service of the Earth system is lost? 1

(a) water filtration and flood control **(b)** ozone
(c) photosynthesis **(d)** gravity

6. In plant cells, the nucleus is pushed to the periphery by the: 1

(a) ER **(b)** chloroplast
(c) large central vacuole **(d)** Golgi

7. A smooth muscle cell has: 1

(a) many nuclei **(b)** branching
(c) a single nucleus and spindle shape **(d)** striations

8. The binomial name 'Canis lupus' and 'Canis familiaris' show that the two: 1

(a) belong to the same species
(b) belong to different families
(c) belong to different genera
(d) belong to the same genus but different species

9. The grafting is successful when the: 1

(a) cambia of stock and scion are in contact **(b)** roots are removed
(c) leaves are cut **(d)** xylems are far apart

10. The stratosphere is warmer at the top due to the: 1

(a) volcanoes **(b)** ground heating
(c) absorption of UV by ozone **(d)** oceans

11. Assertion (A): Mitochondria are called the powerhouse of the cell. Reason (R): They release energy from food. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. The impulse in a neuron travels from: 1

(a) cell body to dendrite **(b)** dendrite to axon
(c) axon to dendrite **(d)** axon to cell body

13. A cell with a cell wall, chloroplast and nucleus belongs to: 1

(a) Monera **(b)** Fungi **(c)** Plantae **(d)** Animalia

14. Seeds dispersed by animals have: 1

(a) air bladders **(b)** hooks or fleshy fruit
(c) wings **(d)** no coat

15. Assertion (A): The natural greenhouse effect is essential for life on Earth. Reason (R): Without it the Earth would be too cold to support life. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. How can you show that plant cells are plasmolysed in salt solution under a microscope? 2

17. What is the effect on a plant if all meristems are destroyed? 2

18. Why do the chordates have a dorsal hollow nerve cord and invertebrates a ventral solid one? 2

19. State two modern methods of assisted reproduction and their purpose. 2

20. Why does the stratospheric ozone absorb UV but not visible light? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Describe how energy flows in the Earth system, from the Sun to the heat radiated to space, taking 100 units reaching the Earth with 30 units reflected. 3

22. Explain the differences between the cell division in the root tip and in the testis. 3

23. A student observed a section showing many xylem vessels in a ring and phloem outside it with a cambium layer between. Identify the plant organ and describe the arrangement. 3

24. A newly discovered organism is unicellular, has a cell wall, a nucleus, no chloroplasts and absorbs food from dead matter. Name its kingdom and give reasons. 3

OR

Compare the reproductive strategies of a frog, a hen and a human as to number of eggs, parental care and survival of young. 3

25. Calculate the energy and CO2 saved if a school replaces 100 tube lights of 40 W by 20 W LED tubes used 8 hours a day for 250 days (0.8 kg per kWh). 3

OR

Why are chloroplasts called the kitchens of the cell? Give three points. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Discuss the causes and prevention of infertility and the techniques of assisted reproduction including IVF, with the social aspects. 5

OR

Explain how deforestation affects the water cycle, the carbon cycle, the albedo and the biodiversity in a region with examples from the Amazon or the Western Ghats. 5

27. Explain the structure and function of a prokaryotic cell and a eukaryotic cell and say why viruses are not considered cells. 5

OR

Describe the structure of the dicot stem and the monocot stem, and the differences in the arrangement of the vascular bundles and the cambium. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND